

ERFAN KHAYYATI

Erfan.Khayyati@usu.edu |  |  | +1 435 962 0531
Logan, UT, USA

Summary

Computer Science Ph.D. student at Utah State University with an M.S. in Computer Science, working in computer vision and vision-language models (VLMs). My recent research experience includes biological video analysis, deep learning model evaluation, and dataset development. I am interested in reliable AI for visual, textual, and multimodal data across scientific and industrial applications.

Education

Utah State University , Logan, UT, USA	2024 – Present
Ph.D. in Computer Science	
M.S. in Computer Science	Spring 2026
- GPA: 3.95/4.00 - M.S. Thesis: Viability Assessment of Bovine Embryos: A Public Dataset and Deep Learning Baselines - Selected Graduate Coursework: Computer Vision, Natural Language Processing, Artificial Neural Networks	
Azarbaijan Shahid Madani University , Tabriz, Iran	
B.Sc. in Computer Science	2019 – 2023
- GPA: 3.49/4.00; Last Two Years GPA: 3.96/4.00; Ranked 1st - B.Sc. Project: Binary Gun Image Classification using CNNs and transfer learning	

Research Interests

- Computer Vision and Video Understanding
- Multimodal Learning and Vision-Language Models (VLMs)
- Medical and Biological Image/Video Analysis
- Large Language Models (LLMs)
- Generative AI and Diffusion Models
- Machine Learning and Deep Learning

Research Experience

Research Assistant , Utah State University	2024 – Present
Bovine Embryo Video Analysis and Viability Prediction	
- Curated and analyzed a bovine embryo time-lapse video dataset with IVF/SCNT samples. - Developed preprocessing pipelines for frame extraction, ROI cropping, resizing, and normalization. - Designed evaluation protocols for six-class classification, outcome-only classification, and embryo-origin generalization. - Benchmarked CNN–RNN, 3D CNN, and video transformer architectures. - Proposed and evaluated a ResNet152-based Hierarchical Chunk-Attention architecture.	
Research Assistant , Azarbaijan Shahid Madani University	2022 – 2023
Gun Image Classification Using Deep Learning	
- Built and cleaned a binary gun image dataset using web scraping and preprocessing. - Trained CNN and transfer learning models, fine-tuned Xception achieved 84% test accuracy.	

Publications / Manuscripts

Erfan Khayyati, et al. “*BovEmbryo: A Time-Lapse Video Dataset and Benchmark for Full-Development Bovine Embryo Viability Assessment*” Submitted to **NeurIPS 2026 Evaluations & Datasets Track**.

Teaching Experience

Teaching Assistant – Utah State University	2024 – Present
- Courses: Graduate Computer Vision, Advanced Data Mining, Algorithms & Data Structures, Foundations of CS. - Graded assignments, exams, and projects; provided technical feedback and mentored students. - Delivered guest lectures on CNN architectures and OpenCV-based computer vision programming.	
Teaching Assistant – Azarbaijan Shahid Madani University	2021 – 2023
Assisted in Deep Learning, Operating Systems, and Advanced Programming courses.	

Selected Academic Projects

- Automatic Short-Answer Grading with LLMs:** Evaluated GPT-4o/GPT-5.2 on the SciEntsBank dataset.
- Skin Cancer Image Classification:** Fine-tuned CNN models including ResNet50 and Xception, on the ISIC dataset.
- Breast Cancer Image Segmentation:** Applied seeded region growing on the MIAS dataset.
- Weakly Supervised Sentiment Analysis:** Evaluated transformer/LLM methods on VAPOR Twitter data.
- Gun Image Classification:** Built a CNN/transfer-learning pipeline on a web-scraped dataset.
- Deep Learning and Advanced Computer Vision Coursework** (Coursera).

Skills

- Programming: Python, JavaScript, C++,
- Deep Learning Frameworks: PyTorch, TensorFlow, Keras
- Machine Learning and Data Science: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- Computer Vision: Image classification, object detection, semantic segmentation, medical imaging, video understanding
- NLP and LLMs: Transformers, text classification, prompt-based evaluation, LLM benchmarking
- Representative Architectures: CNNs, RNNs, Vision Transformers, VLMs, Mamba-based models, diffusion models
- Tools and Platforms: OpenCV, Git/GitHub, Linux, Jupyter Notebook, Google Colab, VS Code

Awards

- Outstanding Computer Science Graduate Teaching Assistant**, Utah State University, 2026.
- Graduate Teaching Assistantship**, School of Computing, Utah State University.
- Summer Graduate Research Assistantship**, Utah State University, 2024–2028.
- Second-Ranked Applicant for Ph.D. Admission**, Utah State University, 2024.
- Ranked 1st in B.Sc. Computer Science Cohort**, Azarbaijan Shahid Madani University.
- Full Four-Year Undergraduate Scholarship**, Azarbaijan Shahid Madani University.

Certificates

- Machine Learning** by Stanford University (Coursera).
- Deep Learning Specialization** by DeepLearning.ai (Coursera).
- Advanced Computer Vision with TensorFlow** by DeepLearning.ai (Coursera).

Professional Experience

Web Development Intern, Fanavaran Tose’e Iranian Ista, Urmia, Iran	Summer 2021
Graphic Designer, Tandis Photography Studio Projects, Urmia, Iran	Summer 2019

Languages

- English:** Fluent, proficiency requirement waived based on U.S. M.S. degree.
- Persian (Native), Turkish (Mother tongue).

References

Available upon request.